Reg. No. | | | | | | | | | | |

[14AE62]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE–2019

FLIGHT MECHANICS

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any five full questions selecting at least two full questions from each part.

PART-A

1.	a.	Derive an expression for determining pressure, temperature and density for the isothermal and gradient regions of International Standard Atmosphere (ISA).	06
	b.	Classify and explain the aircrafts based on the position of their wings with a neat sketch for each case.	05
	c.	Briefly explain the various types of airspeed.	03
2.	a.	List and explain the different types of drag of an aircraft in flight.	09
	b.	Differentiate between a propeller driven aircraft and jet propelled aircraft on the basis of their range and endurance. Also write an expression for each.	05
3.	a.	Describe the Aerodynamic, Wind and Stability axis system with a suitable diagram.	06
	b.	Perform and demonstrate a yaw transform from an initial axis system to a final axis system.	08
4.	a.	Explain the importance of Limit load factor and Ultimate limit load factor in a V-n diagram using suitable diagram.	10
	b.	Derive an expression for specific energy of an aircraft. Consider an aircraft flying at an altitude of 30,000 feet with a velocity of 540 miles per hour. Calculate its specific energy.	04

PART - B

5.	a.	Explain the significance of the stick-free maneuver point with a suitable expression supporting the same.	09
	b.	List down the steps for determining the maneuver point of an aircraft.	05
6.	a.	What are the different types of dynamic stability encountered by an aircraft in flight? Support your classification with neatly labeled diagrams	08
	b.	How does the wing dihedral help in ensuring the lateral stability of an aircraft?	06

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7. **a.** Explain and highlight the advantages and disadvantages of a Canard design with respect to a conventional tail design. How does this design contribute towards aircraft stability? **08**

b. Using proper diagrams, explain briefly about the primary control surfaces responsible for the longitudinal, lateral and directional controls of an aircraft? What is meant by control of an aircraft? **06**

8. **a.** What is aerodynamic balancing? Explain different methods for ensuring that the aircraft is aerodynamically balanced. **07**

b. How are balance tabs different from fixed tabs? Explain each case with a suitable diagram. **07**